

$$\int (10 - 6x + 15x^2) dx \quad \int 30 dx$$

$$10x + C_1 - 6 \int x dx$$

$$\frac{6x^2}{2} \quad \frac{6x^{1+1}}{2} \quad 3x^2$$

730 Nizam Block

430

$$x^{-1/3}$$

$$\int 6x dx$$

$$\int \frac{a}{bx^n} dx$$

$$\frac{a}{b} \int \frac{1}{x^n} dx$$

$$\frac{a}{b} \frac{x^{-n+1}}{-n+1}$$

$$f'(x) = 9x^2 - 4x + 2$$

$$MC = 8x + 20000$$

$$C(x) = \$20000$$

$$3(3x)^{3-1} - 2(2x)^{2-1} + C$$

$$\frac{1}{x}$$

$$8x + 800$$

$$x = 40$$

$$f'(x) = 10 - x + x^2$$

$$4x^2 + 800x + C$$

$$15x^2$$

$$x^2 + x^2 - x + 10$$

$$-2+1$$

$$x(4x + 800)$$

$$\frac{15x^{2+1}}{2+1}$$

$$x^3 \quad \frac{1}{2} \quad \frac{1}{3}x^3$$

$$\frac{1}{3}x^{3-1} \quad \frac{1}{3}(3)x^{3-1}$$

$C(x)$ = total cost function

$$6x \Rightarrow \frac{6x^{1+1}}{2} \quad \frac{15x^3}{3} \quad \frac{15}{3}x^3$$

$$R'(x) = 40000 - 4x$$

$$\int b\sqrt{x} dx$$

$$\int k dx = kx + C$$

$$R=0 \text{ when } x=0$$

$$\int (x)^{1/b} dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

power rule

$$20000x^2$$

$$\frac{x^{1/b}}{1/b+1} + C$$

$$4 \quad 55x^2$$

$$-4x \Rightarrow 2(-2x^{2-1})$$

$$\frac{1}{1/b+1}$$

$$\int kf(x) dx = k \int f(x) dx$$

$$\frac{55x^{2+1}}{2+1}$$

$$-4x$$

$$\int 60 dx = 60x + C$$

$$\int [f(x) + g(x)] dx =$$

$$\int -25 dx = -25x + C$$

$$8x^1$$

$$\int \frac{dx}{8}$$

$$\frac{1}{8}x + C$$

$$\frac{8x^{1+1}}{1+1} \quad \frac{8x^2}{2}$$

$$3(x^1) \Rightarrow 3 \frac{x^{1+1}}{1+1}$$

$$3 \frac{x^2}{2}$$

$$\frac{3}{2}x^2$$

$$\int \left[\frac{x}{2} + \frac{1}{4} \right] dx$$

$$\frac{2}{x^3}$$

$$2x^{-3}$$

$$\int \frac{x}{2} dx + \int \frac{1}{4} dx$$

$$\int (10 - 6x - 15x^2) dx$$

$$\frac{2x^{-3+1}}{-3+1}$$

$$\Rightarrow \int 10 dx - \int 6x dx - 15 \int x^2 dx$$

$$\int \frac{a}{bx^n} dx$$

$$b\sqrt{x}$$

$$x^{1/2}$$

$$\frac{1}{x^2} + \frac{2}{x^3}$$

$$\Rightarrow \frac{a}{b} \int \frac{1}{x^n} dx$$

$$x^{-b}$$

$$\Rightarrow x^{-2} + 2(x)^{-3}$$

$$\frac{a}{b} x^{-n} \Rightarrow$$

$$\frac{x^{-1}}{-3} + \frac{-6x^{-2}}{-2}$$

$$x^{-2} + 2x^{-3}$$

$$\frac{x^{-2+1}}{-3} + \frac{-6x^{-2}}{-2}$$

$$\frac{1}{x} dx = \ln x + c$$

$$e^x dx = e^x + c$$

$$\frac{f'(x)}{f(x)}$$

$$\frac{(x+20)^{5+1}}{5+1} + c \quad [f(x)]^5$$

$$\int (x-30)^{1/2} dx$$

$$[f(x)]^{1/2} \times f'(x) dx$$

$$\Rightarrow \frac{(x-30)^{1/2+1}}{1/2+1}$$

$$10 - \frac{x}{2} \Rightarrow 10 - x^{-2}$$

$$f(x) = 10 - \frac{x}{2}$$

$$\int \left[10 - \frac{x}{2}\right]^4 dx$$

$$\int \frac{1}{2} [10 - x/2]^4$$

$$\Rightarrow \frac{(10 - x/2)^4}{4+1} \times \frac{1}{2}$$

$$\int \frac{1}{2} [10 - \frac{x}{2}]^4 \Rightarrow \frac{(10 - x/2)^4}{4+1} \times \frac{1}{2}$$

$$\frac{(3x-10)^4}{4}$$

$$3x^4 +$$

$$\int \sqrt{8+2x} \cdot 6 dx =$$

$$f(x) = x^3 + 1 \quad f' = 3x^2$$

$$\int (8+2x)^{1/2} \cdot 6 dx$$

$$(x^3+1)$$

$$\int e^{x-8} dx$$

$$2xe^{x^2}$$

$$f(x) = x-8, \quad f'(x) = 1$$

$$\int e^{x-8} dx \quad \int e^{x-8} \cdot 1 dx$$

$$\int \frac{x^3-1}{x^4-4x} \Rightarrow$$

$$e^{x-8} + c$$

7

$$f'(x) = 4x^3 - 4 \quad 4(x^3-1)$$

Ch #10 Linear Programming

10.3

F **1** **2** **3** Cost per oz (\$)

1 50 20 10 0.1

2 30 10 50 0.15

3 20 30 50 0.12

MDR 290 200 210

$$Z = 0.1x_1 + 0.15x_2 + 0.12x_3$$

$$50x_1 + 30x_2 + 20x_3 \geq 290$$

$$20x_1 + 10x_2 + 30x_3 \geq 200$$

$$10x_1 + 50x_2 + 20x_3 \geq 210$$

Wednesday

After Mid-1

06/03/2024

Ch 11 Simplex Method pg 465

Requirements of Simplex Method

Everything must be positive

$$2x_1 + 3x_2 \leq 50$$

$$4x_1 + 2x_2 \leq 60$$

$$x_1 = 10$$

$$x_2 = 20$$

Maximize

Simple Method

$$Z = 5x_1 + 6x_2$$

(const $\leq$)

S_1, S_2 is select variable

$$2x_1 + 3x_2 \leq 50$$

$$2x_1 + 3x_2 + S_1 = 50$$

$$4x_1 + 2x_2 \leq 60$$

$$4x_1 + 2x_2 + S_2 = 60$$

(const $\geq$)

$$x_1 + x_2 \geq 25$$

$$3x_1 + 2x_2 \leq 120$$

(const $\geq$)

$$x_1 + x_2$$

$$4x_1 + 6x_2 \leq 260$$

$$x_1 + x_2 \geq 25$$

$$x_1, x_2 \geq 0$$

$$x_1 + x_2 - E_1 + A_1 = 25$$

$$Z - 5x_1 - 6x_2 - 0S_1 - 0S_2 = 0$$

$$x_1 + x_2 \leq 100 \Rightarrow x_1 + x_2 + S_1 = 100$$

$$3x_1 + 2x_2 + S_1 = 120$$

$$2x_1 + 3x_2 \geq 40 \Rightarrow 2x_1 + 3x_2 - E_2 + A_2 = 40$$

$$4x_1 + 6x_2 + S_2 = 260$$

$$x_1 - 2x_2 = 25 \Rightarrow x_1 - 2x_2 + A_3 = 25$$

Maximize

$$Z = 5x_1 + 6x_2$$

$$\text{Subject to } 3x_1 + 2x_2 \leq 120 \Rightarrow 3x_1 + 2x_2 + S_1 = 120$$

$$4x_1 + 6x_2 \leq 260 \Rightarrow 4x_1 + 6x_2 + S_2 = 260$$

$$x_1, x_2 \geq 0 \Rightarrow x_1, x_2, S_1, S_2 \geq 0$$

$$\text{Solution } Z = 5x_1 + 6x_2 + 0S_1 + 0S_2 \Rightarrow Z - 5x_1 - 6x_2 - 0S_1 - 0S_2 = 0$$

$$\text{Base variable } \begin{bmatrix} x_1 & x_2 & S_1 & S_2 \end{bmatrix}$$

$$S_1 \quad 1 \quad -5 \quad -6 \quad 0 \quad 0$$

$$S_2 \quad 0 \quad 3 \quad 2 \quad 1 \quad 0$$

$$0 \quad 4 \quad 6 \quad 0 \quad 1$$

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Monday

March 11, 2024

Ch. 11 Simplex Method

Ex. 11 Maximize

$$Z = 2x_1 + 2x_2 + 8x_3$$

Subject to

$$2x_1 + 8x_2 + x_3 \leq 100$$

$$x_1 - 2x_2 + 5x_3 \leq 80$$

$$10x_1 + 5x_2 + 4x_3 \leq 300$$

$$x_1, x_2, x_3 \geq 0$$

Base variable	Z	x_1	x_2	x_3	s_1	s_2	s_3	
s_1	0	1	-2	-8	0	0	0	100 (0)
s_2	0	2	2	1	1	0	0	100 (1)
s_3	0	-2	5	5	0	1	0	80 (2)
s_3	0	10	5	4	0	0	1	300 (3)
	0	1	1	1/2	1/2	0	0	50
	1	10	$\frac{100-50}{2} = 25$	$\frac{300-50}{2} = 125$	$\frac{300-50}{2} = 125$	$\frac{80+50}{2} = 65$	$\frac{80+100}{2} = 90$	
	0	3	0	6	6	0	0	180
	0	5	0	3/2	-5/2	0	1	50

Simplex Method

$$Z = 2x_1 + 12x_2 + 8x_3 + 0s_1 + 0s_2 + 0s_3$$

$$2x_1 + 2x_2 + x_3 + s_1 = 100$$

$$x_1 - 2x_2 + 5x_3 + s_2 = 80$$

$$10x_1 + 5x_2 + 4x_3 + s_3 = 300$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	$R_2 + 12R_1$
	1	10	0	-2	6	0	0	600
x_2	0	1	1	1/2	1/2	0	0	50
s_2	0	3	0	6	1	1	0	130
s_3	0	5	0	3/2	-6/2	0	1	50

$$\begin{matrix} Z \\ s_1 \\ s_2 \\ s_3 \end{matrix} \begin{pmatrix} -12 \\ 2 \\ -2 \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$R_0 + 12R_1$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	1	0	0	19/3	1/3	0	660
x_2	0	3/4	1	0	5/12	-1/12	0	35
x_3	0	1/2	0	1	1/6	1/6	0	30
s_3	0	17/4						

Wednesday

13/03/2024

گسټی کسټی ما ټول ټول ټول

Slack variables are basic variable always

Artificial variables are basic

pg 493 Maximize the value of

$Z = 8x_1 + 6x_2$	BV	Z	x_1	x_2	E_1	A_1	S_1	
Subject to		1	-8	-6	0	M	0	0
$2x_1 + x_2 \geq 10$	A_1	0	2	1	-1	1	0	$10 + 10M = 5$
$3x_1 + 8x_2 \geq 96$	S_2	0	3	8	0	0	1	$96 + 96M = 96$
$x_1, x_2 \geq 0$								
$Z = 8x_1 + 6x_2 +$								
$Z - 8x_1 - 6x_2 - 0E_1 + MA_1 - 0S_2 = 0$								
$2x_1 + x_2 - E_1 + A_1 = 10$		Z						
$3x_1 + 8x_2 + S_2 = 96$		1	-8-2M	-6M	M	0	0	$-10M$
$x_1, x_2, E_1, A_1, S_2 \geq 0$	A_1	0	2	1	-1	1	0	10
	S_1	0	3	8	0	0	1	96

	Z	x_1	x_2	R_1	R_2	S_2	
x_1	0	1	$1/2$	$-3/2$	$1/2$	0	5
S_2	0	0	$13/2$	$3/2$	$-3/2$	1	
							$\begin{pmatrix} -8-2M \\ 2 \\ 3 \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$
							$M = 8 - 2M$
							$-6 - M + 1/2(8 + 2M)$
							$-6 - M + 1/2(8 + 2M)$
							$-6 - M + 4 + M$
							$M - 1/2(8 + 2M)$
							$M - 4 - M$
							$0 + 1/2(8 + 2M)$
							$-10M + 5(8 + 2M)$
							$4 + M$
							$-10M + 40 + 10M$

Z x_1 A_1 R_1 A_1 / S_2

$(R_1 + 1/2 P_1)$

x_1

$R_0 + 4R_2$

Danish Amjad Alvi

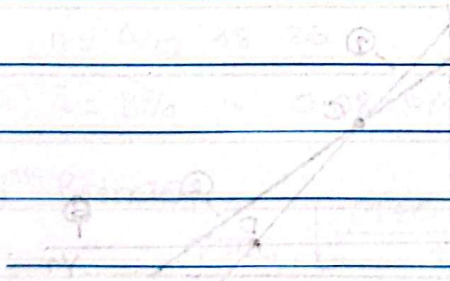
E1 0 0 $13/3$ 1 -1 $3/3$ 54

Floop Mathematician

Physicist $\rightarrow$ Math

Assignment Questions

11.2 10 questions Solve any 5 to maximize



Monday

Business Maths II

18/03/2024

Chapter 11 Fig 11.3

constraint numbers?

- (1) you'll get an idea of constraint from feasible
- (2)
- (3) region
- (4)
- (5) constraints is greater than or less than?

(1) $\geq$ E1 A1 surplus is known as slack variable?

(2) $\leq$ S2 yes it is slack and artificial

(3) $\leq$ S3 CS370) constraint number one which variables are supplement?

(4) $\leq$ S4

(5) $\geq$ E5 A5 CS470) 11 & 14

Basic and Non Basic variables?

Mahad Jawad

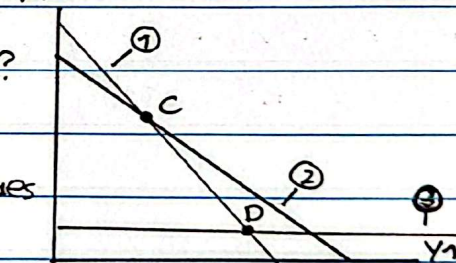
feasible region
less than
(1) S1
(2) S2
(3) E3, A3

Simplex Method? it is a plus minus rule?

Maximization Rule?

Minimization Rule no 1 is applied in slack variables

Rule?



$$\frac{PRT}{100} P(1+i)^n R = \frac{A}{a_{\overline{n}|i}}$$

$$R \left[\frac{(1+i)^n - 1}{i} \right]$$

$$A = R \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$A = R \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$



Ch#8

$$P(1+i)^n \frac{PRT}{100}$$

Ex#8.3

Duration extends? compound interest increases?

Annuity?

$$P(1+i)^n \frac{PRT}{100}$$

R = Amount of an annuity

n = number of annuity payments

Sn = sum future value of the annuity n periods

$$S_n = R + R(1+i) + R(1+i)^2$$

$$\Rightarrow R(1+i)^{n+1}$$

$$R \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] R \left[\frac{(1+i)^n - 1}{i} \right]$$

Future value formula

$$S_n = R [1 + (1+i) + (1+i)^2 + (1+i)^3 + \dots + (1+i)^n] \frac{(1+i)^n - 1}{i} S = R \left[\frac{(1+i)^n - 1}{i} \right]$$

$$S_n = (1+i) + R(1+i)$$

$$S_n = S_n i = R [(1+i) + (1+i)^2 + (1+i)^3 + \dots + (1+i)^n] \frac{(1+i)^n - 1}{(1+i)^{n+1} - 1} 50 \left[\frac{(1+0.02)^6 - 1}{1} \right]$$

R(1+i)^n factor?

$$S_n = R(1+i)^n - R$$

کسی کسی

$$i S_n = R [(1+i)^n - 1]$$

$$S_n = 0.02$$

Ans 3)

$$S = \$750$$

$$0.08 \times 1/4 \Rightarrow 0.02$$

$$24 \times 50$$

$$n = 6/12 \quad 18 \quad 36$$

$$n = 2 \times 18 = 36$$

$$i = 8\%$$

$$0.08 \times 6/12 \Rightarrow 0.04$$

$$0.04$$

18th Birthday

$$\$58198$$

$$\$750 (1+0.04)^{18} \Rightarrow 1519.37$$

$$\$750 \left[\frac{(1+0.04)^{18} - 1}{1} \right]$$

$$(1+0.04) + 750 (1+0.04)$$

$$769.362$$

$$2 \times 18 \Rightarrow 781.04$$

$$750 \left[\frac{(1+0.04)^{36} - 1}{-1} \right]$$

$$i = 0.04$$

$$1577$$

$$\$250000 = R \left[\frac{(1+0.025)^8 - 1}{0.025} \right] \Rightarrow 8.7361159$$

$$28616.83646$$

Per year compounded semiannually? Divided by 2

compounded semiannually? Not divided
by 2



Quiz Ch 8?

8.3
8.4

h
a 13 20
28

Business Math - II

8.4 Annuities at their Present value

Question No. 10

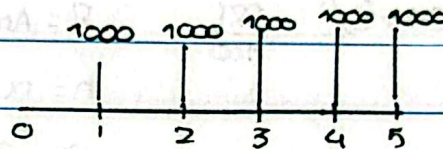
Minimize $Z = 100x_1 + 75x_2$

$$x_1 + x_2 \geq 200$$

$$x_2 \geq 100$$

$$x_1 \geq 80$$

$$x_1, x_2 \geq 0$$



$$P(1+i)^n$$

$$1000(1+0.1)^{-1} = 909$$

$$1000(1+0.1)^{-2} = 826$$

$$1000(1+0.3)^{-3} = 751$$

Diff of interest rate

2486 → Market value

4:03 funny way
5:03 4th guy
5:07 ruko jorra
5:15 intense

Ans 1

$$\$25000 \left[\frac{(1+0.09)^{10} - 1}{0.09(1+0.09)^{10}} \right]$$

$$A = R \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

S

$$\Rightarrow 160441$$

Ans 3: $A = 25?$

$A = 7?$

$R = \$10000$ $n = 25$

$n = 10$

6 months

$$0.1/2 = 0.05$$

$R = \$25000$

0.7854517926

10% 1-year compounded semiannually

$i = 9\% \sim 0.09$

$$10000 \left[\frac{(1+0.05)^{25} - 1}{0.05(1+0.05)^{25}} \right] \Rightarrow 140939$$

$$A = ? \quad \$25000 \left[\frac{(1+0.045)^8 - 1}{0.045(1+0.045)^8} \right]$$

$$R = \$25000 \Rightarrow 164897.73$$

$i = 0.0912$

$n = 8$

$$R = \frac{A}{n \cdot i}$$

$n = 20$

lump sum amount \$50000

1 year from the time of death

$$\$50000 = R \left[\frac{(1+0.08)^{20} - 1}{1(1+0.08)^{20}} \right] \quad i = 0.08 \quad n = 20$$

$$50000 = R \cdot 0.786$$

$A = 50000$

50000

$$63657.63059$$

$$\frac{P(1+i)^n}{\frac{PRT}{100}} A = R \left[\frac{(1+i)^n - 1}{1(1+i)^n} \right]$$

$$Z = 100x + 75x^2$$

$$x_1 + x_2 \geq 200$$

$$x_2 \geq 100$$

$$x_1 \geq 80$$

$$x_1, x_2 \geq 0$$

$$Z = f(x, y)$$

$$f(x, y) = -10xy^3$$

$$Z = 25 - x^2 + y^2$$

$$f_x = -10x(3y^2)$$

$$Z = 100x - 75x^2 - 0E_1 - 0E_2 - 0E_3$$

$$Z_x = 0 - 2x + y^2$$

$$-10x(3y^2)$$

$$-MA_1 - MA_2 - MA_3$$

$$\frac{\partial Z}{\partial y} = 0 - 0 + 2y \Rightarrow$$

$$\frac{P(1+i)^n - 1}{i}$$

compounded semi annually?

$$\frac{R(1+i)^n - 1}{i(1+i)^n}$$

per year compounded semi annually? Divided by 2

$$\frac{4000}{1500}$$

500 PKR ✓

$$Z = 6x_1 + 4x_2$$

2500 outstanding dues? ✓

$$6x_1 + 10x_2 \leq 90$$

$$12x_1 + 8x_2 \leq 96$$

$$x_1, x_2 \geq 0$$

$$6x_1 + 10x_2 = 90$$

$$12x_1 + 8x_2 = 96$$

$$Z = 6x_1 + 4x_2$$

$$x_1, x_2, s_1, s_2$$

$$6x_1 + 10x_2 \leq 90$$

$$Z = 6x_1 - 4x_2$$

$$Z = 7x_1 + 2x_2 + 5x_3$$

$$x_1 + 3x_2 + x_3 = 35$$

Monday

15/04/2024

Applications of Bivariable

Example 18 Maximize?

$$z = 50000x + 40000y - 10x^2 - 20y^2 - 10xy$$

So,

$$f_x = 50000 - 20x - 10y$$

$$f_y = 40000 - 40y - 10x$$

So, Multiply f_y by -2 & solve simultaneously (Adding)

$$20x + 10y = 50000$$

$$-20x - 80y = -80000 \quad | y = 428.57$$

$$-70y = -30000$$

Now, Putting y in $f_x \rightarrow$ Now

$$20x + 10(428.57) = 50000$$

$$| x = 2285.72 |$$

$$f(2285.72, 428.57) = 50000(2285.72) + 40000(428.57)$$

$$- 10(2285.72)^2 - 20(428.57)^2 - 10(2285.72)(428.57)$$

$$f(2285.72, 428.57) = 65,714,296 \text{ units}$$

C.P occurs at $f(2285.72, 428.57, 65,714,296)$

$$f_{xx} = -20$$

$$f_{xy} = -10$$

$$f_{yy} = -40$$

$$f_{yx} = -10$$

When $D \times D$ (test failed)

$$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$$

$$D = (-20) \cdot (-40) - (-10)^2$$

$$D = 700 > 0$$

So,

$D > 0$ and f_{xx} and f_{yy} are negative,

So we conclude that sales are maximum

at 65 714 296 units annually at $x = 2285.57$

Example 19 Total Revenue Maximization

$$q_1 = 150 - 2P_1 - P_2$$

$$q_2 = 200 - P_1 - 3P_2$$

$$\text{Q3 } q_1 = 110 - 4P_1 - P_2$$

$$q_2 = 90 - 2P_1 - 3P_2$$

First, we have to find the revenue functions

$$R = P_1 q_1 + P_2 q_2$$

$$R = 150P_1 - 2P_1^2 - 2P_1P_2 + 200P_2 - 3P_2^2$$

$$R_{P_1} = 150 - 4P_1 - 2P_2 \quad P_1 = 25$$

$$R_{P_2} = 200 - 6P_2 - 2P_1 \quad P_2 = 25$$

$$D = 20 > 0, \quad R_{P_1 P_1} = -4 \quad R_{P_1 P_2} = -6$$

$$R_{P_1 P_2} = -2$$

$$\boxed{R = 4375} \text{ Maximum}$$

Chapter 18 INTEGRATION / ANTI DERIVATIVES

$$f(x) = 4x + 5$$

$$f(x) = 4$$

$$f(x) = 4x + C \rightarrow \text{constant? No idea if it's constant or not?}$$

Application of integration? Find C

Normally it is arbitrary

$$f(x) = 4x + 3$$

$$\int f(x) dx = \int (4x + 3) dx$$

$$= \int 4x dx + \int 3 dx$$

$$= 4 \int x dx + 3 \int dx$$

$$4 \frac{x^{1+1}}{1+1} + 3x + C$$

$$\frac{4x^2}{2} + 3x + C$$

$$2x^2 + 3x + C$$

$$\int (2x^2 - 3x + 5) dx$$

$$2 \int x^2 dx - 3 \int x dx + 5 \int dx$$

$$2 \frac{x^{2+1}}{2+1} - \frac{3x^{1+1}}{1+1} + 5x + C$$

$$\frac{2}{3}x^3 - \frac{3x^2}{2} + 5x + C$$

$$MC = x + 100$$

$$TC = 40000 \text{ when } x = 100$$

Cost Function?

$$\int MC dx = \int (x + 100) dx$$

Ex 6 $MR = 50000 - x$

$R = 0$ when $x = 0$

$R(x) = ?$

$$\frac{x^2}{2} + 100x + C$$

$$C(100) = \frac{(100)^2}{2} + 100(100) + C$$

$$C = 25000$$

Power Rule

تک باقی می‌ماند integrate تک باقی می‌ماند differenti

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$(11) \int \left[\frac{x}{3} - \frac{1}{4} \right] dx$$

$$\int \frac{x}{3} dx - \int \frac{1}{4} dx$$

$$\frac{1}{3} \cdot \frac{x^2}{2} - \frac{1}{4} x + c$$

$$\frac{x^2}{6} - \frac{x}{4} + c$$

$$(27) \int \frac{a}{bx^n} dx$$

$$\frac{a}{b} \int x^{-n} dx$$

$$\frac{a}{b} \frac{x^{-n+1}}{-n+1} + c$$

$$(6) \int \frac{x}{2} dx$$

$$\frac{1}{2} \int x dx$$

$$\frac{1}{2} \cdot \frac{x^2}{2} + c$$

$$\frac{x^2}{4} + c \checkmark$$

$$(24) \int \frac{20}{3\sqrt{x^2}} dx$$

$$\frac{1}{2} \int \frac{2(a-bx)}{2ax-bx^2} dx = \frac{1}{2} \ln |2ax - bx^2| + c$$

$$20 \int \frac{1}{(x^2)^{1/3}} dx \Rightarrow 20 \frac{x^{1/3}}{1/3} + c$$

$$20 \int x^{-2/3} dx$$

$$20 x^{-2/3+1} + c$$

$$20 \frac{x^{-2/3+1}}{-2/3+1} + c$$

$$\frac{1/3}{60} x^{1/3} + c$$

$$60 \sqrt[3]{x} + c$$

$$(9) \int (x^2+3)^4 2x dx$$

$$\frac{(x^2+3)^5}{5} + c$$

$$(25) \int (x^3+3x^4)(3x+12x^2) dx$$

LOG Rule

$$\int \frac{1}{x} dx = \ln x + c$$

$$\int e^x dx = e^x + c$$

$$(5) \int \left[\frac{x}{3} + 15 \right]^2 \frac{1}{3} dx$$

$$\frac{\left[\frac{x}{3} + 15 \right]^3}{3} + c \checkmark$$

$$(27) \int e^{x^2} dx$$

$$(30) \int 2xe^{x^2} dx$$

$$\int e^{x^2} 2x dx$$

$$e^{x^2} + c$$

$$(1) \int (x+20)^5 dx$$

$$\frac{(x+20)^6}{6} + c \checkmark$$

$$(7) \int (3x-10)^3 \cdot x dx$$

$$(40) \int \frac{a-bx}{2ax-bx^2} dx$$

$$\frac{1}{2} \int \frac{2x(a-bx)}{2ax-bx^2} dx = \frac{1}{2} \ln |2ax - bx^2| + c$$

$$\int u v dx$$

$$u \int v dx - \int (u \frac{d}{dx} \int v dx) \cdot u dx$$

$$(2) \int 5x e^x dx$$

$$(5) \int x e^{ax} dx$$

$$Ex // \int \frac{x}{x^2+1} dx$$

Is this pen working fine or not?

Integ by parts

$$(32) \int e^{-2x} dx$$

$$= x e^x - \int e^x (1) dx$$

$$= \frac{e^{-2x}}{-2} + c \checkmark$$

$$= x e^x - \int e^x dx$$

$$x e^x - e^x + c \checkmark$$

Wednesday

24/04/2024

Lateral Integration

power rule $\frac{n+1}{n+1}$

$$(4) \int x \sqrt{x+1} dx$$

Integrate by parts

$$\int \sqrt{x+1} dx \Rightarrow \frac{(x+1)^{3/2}}{3/2}$$

$$x \cdot \frac{(x+1)^{3/2}}{3/2} - \int \frac{(x+1)^{3/2}}{3/2} (1) dx$$

$$\frac{2}{3} x (x+1)^{3/2} - \frac{2}{3} \int (x+1)^{3/2} dx$$

$$\frac{2}{3} x (x+1)^{3/2} - \frac{2}{3} \frac{(x+1)^{5/2}}{5/2} + c$$

$$\Rightarrow \frac{2}{3} x (x+1)^{3/2} - \frac{4}{15} (x+1)^{5/2} + c$$

$$\begin{aligned} (7) \quad & \int (x+4) \ln x \, dx \\ & \int x \ln x \, dx + \int 4 \ln x \, dx \\ \Rightarrow & \ln x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx + 4 \int 1 \cdot \ln x \, dx \\ \Rightarrow & \ln x \cdot \frac{x^2}{2} - \frac{1}{2} \int x \, dx + 4 \left[\ln x - \int x \cdot \frac{1}{x} \, dx \right] \\ \Rightarrow & \frac{x^2}{2} \ln x - \frac{1}{2} \cdot \frac{x^2}{2} + 4x \ln x - 4x + C \end{aligned}$$

$$\begin{aligned} (29) \quad & \int x^4 \ln x \, dx \\ & \int \ln x \cdot x^4 \, dx \\ \Rightarrow & \ln x \cdot \frac{x^5}{5} - \int \frac{x^5}{5} \cdot \frac{1}{x} \, dx \\ \Rightarrow & \frac{x^5}{5} - \frac{1}{5} x \int \frac{x^4}{dx} \end{aligned}$$

$$\begin{aligned} (13) \quad & \int \ln x \cdot \frac{1}{x^2} \, dx \\ \Rightarrow & \int \ln x \cdot x^{-2} \, dx \\ & \ln x \cdot \frac{x^{-2+1}}{-2+1} - \int \frac{x^{-1}}{-1} \cdot \frac{1}{x} \, dx \\ \Rightarrow & \ln x \left[-\frac{1}{x} \right] + \int x^{-2} \, dx \\ \Rightarrow & -\frac{\ln x}{x} + \frac{x^{-1}}{-1} + C \\ & -\frac{\ln x}{x} + \frac{x^{-1}}{-1} + C \\ \Rightarrow & -\frac{1}{x} \ln x - \frac{1}{x} + C \end{aligned}$$

$$\begin{aligned} (33) \quad & \int x e^{5x} \, dx \\ & x \cdot \frac{e^{5x}}{5} - \int \frac{e^{5x}}{5} (1) \, dx \\ \Rightarrow & \frac{x e^{5x}}{5} - \frac{1}{5} \int e^{5x} \, dx \\ \Rightarrow & \frac{x e^{5x}}{5} - \frac{1}{5} \int e^{5x} \, dx \\ \Rightarrow & \frac{x e^{5x}}{5} - \frac{1}{5} \cdot \frac{e^{5x}}{5} + C \end{aligned}$$

Ordinary differential equation
partial differential equation

$$x \frac{dy}{dx} + x^2 - 4$$

Degree of equation type $\rightarrow$ order and degree of differential equation

Antiderivative equation Second point degree

order one degree two $\rightarrow$

$$\frac{d^2 y}{dx^2} = x - 5 \Rightarrow$$

$$f'(2) = 4, f(0) = 10$$

$$\frac{d^2 y}{dx^2} = x - 5$$

$$\int \frac{d^2 y}{dx^2} \, dx = \int (x - 5) \, dx$$

$$\frac{dy}{dx} = \frac{x^2}{2} - 5x + C$$

$$4 = \frac{4}{2} - 10 + C$$

$$4 = -8 + C \quad [C = 12]$$

Q. $\frac{dy}{dx} = \frac{x^2}{2} - 5x + C_1$

$y = \frac{x^3}{6} - 5 \cdot \frac{x^2}{2} + C_1 x + C_2$

$10 = 0 - 0 + 0 + C_2$

$C_2 = 10$

G.S General Solution C_1, C_2 only

$y = \frac{x^3}{6} - \frac{5}{2} x^2 + C_1 x + C_2$

(21) $\frac{d^2 y}{dx^2} = e^x + 5$

$\int \frac{d^2 y}{dx^2} dx = \int e^x + 5 dx$

$\frac{dy}{dx} = \frac{x^2}{2} - x \cdot \frac{e^x}{x}$

$\int \frac{dy}{dx} dx = \int e^x + 5x dx + C$

$y = e^x + \frac{5}{2} x^2 + C_1 x + C_2$

Question 1-10 order of degree

11-28 General Solution

29-38 Particular Solution

P.S Particular Solution value of C_1, C_2 in particular solution

$y = \frac{x^3}{6} - \frac{5}{2} x^2 + 12x + 10$

Q # 35

$\frac{d^2 y}{dx^2} = 25 e^{5x}$

$f'(0) = 4$

$f(0) = -2$

$\int \frac{5-x}{x^2+5x+6} dx$

$x(x+5)+6$

$(x+6)(x+5)$

x^2+5x+6

$(x+2)(x+3)$

$$\int_{-1}^1 -8x dx = -8 \int_{-1}^1 x dx$$

$$\frac{x^2}{2} \sim \frac{(1)^2}{2} - \frac{8x^2}{2} \bigg|_{-1}^1 = -8 \left[\frac{x^2}{2} \right]_{-1}^1$$

$$= -8$$



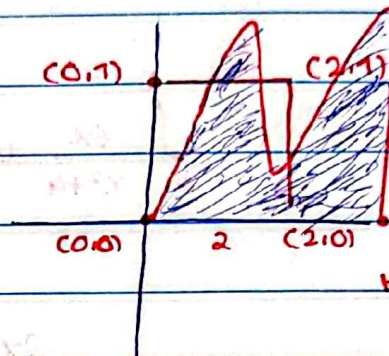
Chapter 19 Definite Integral

Pg 874 questions are very simple

$$\int_a^b f(x) dx = F(x) \bigg|_a^b$$

$$= F(b) - F(a)$$

Area under the curve



If suppose it is an irregular shaped curve then you can find out integration

How to calculate distance?

Final - initial

In case you have to put an upper limit - lower limit

rectangle area = $L \times W$

circle = πr^2 for specific shapes only

① $\int_0^3 x^2 dx$

$$\frac{x^3}{3} \bigg|_0^3 = \frac{3^3}{3} - \frac{0^3}{3}$$

$$9 - 0 = 9$$

$$\int_1^4 (2x^2 - 4x + 5) dx$$

$$\frac{2x^3}{3} - 2x^2 + 5x \bigg|_1^4$$

$$\int_{-2}^1 e^x dx$$

$$\frac{e^x}{1} = e^x \bigg|_{-2}^1$$

$$= e^1 - e^{-2}$$

2.58 No power rule in e^x

$$\frac{2x^3}{3} - 4x + 5x \bigg|_1^4$$

$$\frac{2x^3}{3} - 2x^2 + 5x \bigg|_1^4 = \left[\frac{2(64)}{3} - 2(16) + 20 \right] - \left[\frac{2}{3} - 2 + 5 \right]$$

$$\left[\frac{128}{3} - 12 \right] - \left[\frac{2}{3} + 3 \right] = 27$$

$$\frac{28-36}{3} = \frac{-8}{3}$$

$$92/3 - 11/3 = 81/3 = 27$$

$$\int_0^4 2x(x^2+4)^2 dx \rightarrow \frac{(x^2+4)^3}{3} \bigg|_0^4$$

$$\frac{(20)^3}{3} - \frac{4^3}{3}$$

$$\int_0^2 (x+3)^2 dx = \left. \frac{(x+3)^3}{3} \right|_0^2$$

$$\frac{1}{3} \int_2^3 \frac{3x^2}{x^3-1} dx$$

$$= \frac{1}{3} \ln(x^3-1) \Big|_2^3$$

$$= \frac{1}{3} [\ln 26 - \ln(7)]$$

$$\frac{1}{3} \ln 26/7$$

$$\int_1^4 \frac{3x^2+x}{8x^3+4x^2} dx$$

(50)

$$\int_3^5 \frac{6x}{x^2+4} dx = 3 \int_3^5 \frac{2x}{x^2+4} dx$$

$$\int_{50}^{15} (x-5) dx$$

$$\int_0^3 (g(x)-f(x)) dx$$

$$\frac{2x}{x^2-4} \Rightarrow \int \frac{f'(x)}{f(x)}$$

$$\Rightarrow \frac{2}{x^2-4} = \ln|f(x)|$$

$$\int_0^3 (2x^2+27-0x^2) dx$$

$$\int_0^3 (f_1-f_2)$$

$$2x \quad f'(x) \rightarrow x^2$$

$$\Rightarrow \int_0^3 (-3x^2+27) dx$$

$$-x^3+27x \Big|_0^3$$

$$-27+81=54$$

$$\int_2^5 (f_2-f_1) dx + \frac{x^2}{2} - 5x \Big|_5^{15}$$

$$\int_6^5 (0-f_1) dx \left[\left[\frac{225}{2} - 75 \right] - \left[\frac{25}{2} - 25 \right] \right] + \frac{25}{2}$$

$$f(x) = x^2$$

$$g(x) = -2x^2+27 \quad 40x-x^2=0$$

$$x^2 = -2x^2+27 \quad x(40-x)=0$$

$$3x^2=27 \quad x=0 \quad x=40$$

$$x^2=9$$

until or unless you had dinner for your own
Do not use any harmful device or thing at night

Exercise 19.3

55-250 mm STM Lens Canon Lens

Rectangular Methods

$$\Delta x = \frac{b-a}{n}$$

Good to go for prank shoots

$$\int_0^4 x^3 dx = \frac{x^4}{4} \Big|_0^4 = \frac{4^4}{4} = 0$$

$$(6) \int_1^5 3e^x dx$$

$$\int_0^4 x^3 dx$$

$$\frac{x^4}{4} \Big|_0^4 = \frac{4^4}{4} = 0$$

$$f(x) = x^3 \quad 4^3 \quad 64$$

$$3e^x \Big|_1^3$$

$$a=0, b=4$$

$$\Delta x = \frac{b-a}{n} = \frac{4-0}{4} = 1 \quad 0.1-2-3-4$$

$$\int_4^3 2x(x^2-5)^3 dx$$

$$\int_4^3 (x^2-5)^3 2x dx$$

$$A = \Delta x \left[\frac{v_1 + v_2}{2} \right]$$

$$x_1 = 0.5 \Rightarrow f(0.5) = 0.125$$

$$x_2 = 1.5 \Rightarrow f(1.5) = 3.375$$

$$x_3 = 2.5 \Rightarrow f(2.5) = 15.625$$

$$x_4 = 3.5 \Rightarrow f(3.5) = 42.87$$

Trapezium Rule

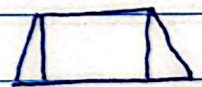
$$\frac{1}{2} (h)(l_1 + l_2)$$

Rectangle
Length x Width
height Length

Area error would reduce

$$\int_a^b f(x) dx \approx f(x_1) \Delta x + f(x_2) \Delta x + \dots + f(x_n) \Delta x$$

$$\Delta x [f(x_1) + f(x_2) + \dots + f(x_n)]$$



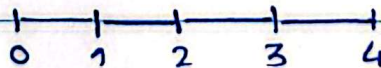
$$\frac{1}{2} (h)(l_1 + l_2)$$

$$\int_a^b x^2 dx$$

$$a=0?$$

$$b=4$$

$$n=4$$



square and add them all up

$$\frac{b-a}{n}$$

$$\frac{4-0}{4} \Delta x = 1$$

divide end point of intervals by 2

$$\frac{0+1}{2} = 0.5$$